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Alfonso Ortega is the James R. Birle Professor of Energy Technology and Associate Dean for Graduate Studies and Research at Villanova University. He received his B.S. in 1976 from The University of Texas-El Paso, and his M.S. and Ph.D. from Stanford University in 1978 and 1986, respectively, all in Mechanical Engineering. At Stanford University, he studied with Prof. Robert J. Moffat in the well-known laboratory for heat transfer and turbulence mechanics. He started his professional career at Sandia National Laboratories in Albuquerque, New Mexico, where he served as a member of the technical staff from 1978 to 1981 and 1986 to 1988, focusing on research in solar and geothermal energy topics. From 1988 to 2005, Professor Ortega was on the faculty of the Department of Aerospace and Mechanical Engineering at The University of Arizona in Tucson. He established the Experimental and Computational Heat Transfer Laboratory in 1988, which he directed until 2005.

From 2004 to 2006, Dr. Ortega was the Program Director for Thermal Transport and Thermal Processing in the Chemical and Transport Systems Division of The National Science Foundation in Arlington, Virginia, managing a $6.5 million program that, under his tenure, awarded in excess of $12 million in grants in emerging areas of thermal science. During 2005-2006, he was the Program Coordinator of the Active Nanostructures and Nanosystems NIRT Program, coordinating the efforts of 15 programs across the engineering directorate, which awarded $35 million in emerging nanoengineering topics. He was the engineering representative to the NSF-wide IGERT graduate research traineeship program and the engineering REU sites program, and initiated and participated in numerous early-career-development workshops aimed at engaging and retaining underrepresented engineering faculty.

While at the NSF, he presented keynote and invited lectures at more than 25 universities and symposia in the U.S., Europe, and Asia. He initiated a unique joint research program with the NIH to engage the engineering community to address issues related to energy balance and obesity, and also initiated a major NSF workshop on Emerging Frontiers in Thermal Transport, with the goal of identifying emerging topics and helping the NSF to define the thermal transport research agenda for the next decade.

Dr. Ortega joined the faculty of Mechanical Engineering at Villanova University as the James R. Birle Professor of Energy Technology in 2006. That same year, he established the Laboratory for Advanced Thermal and Fluid Systems (www.villanova.edu/latfs), which is the focal point for his research in fundamentals and applications of heat transfer in thermal-fluid systems. In 2008, he was named Associate Dean for Graduate Studies and Research. In this position, he has administrative oversight over all graduate programs and research initiatives in the College of Engineering.

Dr. Ortega is a teacher of the science and design of thermal systems, as well as an internationally recognized authority in the cooling of electronic systems, convective and conjugate heat transfer in complex flows, and experimental measurements in the thermal sciences. The Laboratory for Advanced Thermal and Fluid Systems, which he currently directs, is dedicated to fundamental and applied research in the heat transfer and fluid mechanics fundamentals of convective heat transfer in single and two phase flow, especially in problems that arise from the technology of electronics thermal management, gas turbine cooling, and alternative energy technologies.

Dr. Ortega has supervised nearly 40 M.S. and Ph.D. candidates to degree completion, 5 postdoctoral researchers, and more than 70 undergraduate research students. He is the author of over 300 journal and symposia papers, book chapters, and monographs. He has presented numerous invited lectures on these subjects nationally and internationally, and is a frequent short course instructor on the subject. He has held visiting positions at The National Institute of Standards and Technology (NIST) and the Technical University at Delft, The Netherlands (TU-Delft), and has served on international doctoral committees in Sweden, The Netherlands, and Korea. He is currently an Associate Editor of the ASME *Journal of Heat Transfer*, a former Associate Editor of the ASME *Journal of Electronic Packaging* and Guest Associate Editor of the IEEE *Transactions on Components, Packaging, and Manufacturing Technology*.

Dr. Ortega is a Fellow of the ASME (2005) and a member of the Society for Hispanic Professional Engineers (SHPE), the Society for the Advancement of Chicanos and Native Americans in Science (SACNAS), ASEE, and AIAA. He is currently Chair of the ASME Committee on Division Health and a Member at Large of the Systems and Design Group Operating Board. He is a former Chair of the ASME K16 Committee on Heat Transfer in Electronic Equipment (1993-1997) and of the ASME Electronic and Photonic Packaging Division (2001-2002). He was the General Chair of the ASME/IEEE ITHERM Symposium (1994), the IEEE SEMITHERM Symposium (1994), and the 2005 ASME International Electronic and Photonic Packaging Conference (InterPACK 2005).

Dr. Ortega is the recipient of the National Science Foundation Presidential Young Investigator Award (1990), the 2001 ASME Electronic Packaging Division Thermal Management Award, and the 2002 IEEE SEMITHERM Significant Contributor Award. He was awarded a U.S. Army Medal of Commendation to recognize research work of lasting benefit to the U.S. Army (2003). In 2004, Dr. Ortega received the inaugural Distinguished Service Award from Southern Arizona MESA (Math, Engineering, and Science Achievement) for his many contributions to the math and science development of southern Arizona's underrepresented middle and high school students. That award is now named in his honor.

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